NAME: Madhumitha. K 241801145.

STD: AIDS - III DIV: ROLL NO.:

SUBJECT:

YOUVA

INDEX

SR. NO.	DATE	TITLE	PAGE NO.	TEACHER'S SIGN
1.	16.07.26	Basic Computer networking command		BM 6/8
2.	23.7.26	Study of different types of network cables		BM 6/8
3.1	23.7.26	HUB		BM 6/8
3.2	23.7.26	SWITCH		BM 6/8
3.3	23.7.26	HUB & SWITCH		BM 6/8
4.	30/7/26	Situn & Configure LAN using Switch on ethernet		BM 6/8
5.	1/8/26	Packet capture tool.		BM 6/8
6	6/8/26	Hamming code error correction		BM 20/8
7.	6/8/26	Sliding window Protocol		
9.	7/8/26	Flow control at data Link layer using Sliding window protocol		
9.	10/9/26	Cisco Packet Tracker		BM 16/10
10.	10/9/26	Networking using Router		BM 16/10
12.	13/9/26	Chat Server. Java		BM 16/10
11	24/9/26	Routing at Network layer.		BM 16/10
13.	8/10/26	Implement your own Ping Program.		BM 16/10
14.	8/10/26	Write a code using raw sockets to implement Packet sniffing		BM 16/10
*	8/10/26	Wireless LAN.		BM 16/10

DATE

TITLE

PAGE NO.

TEACHER'S SIGN

28/10/26

Analyse Various Types of Servers using webalizer tool.

Date : 16.07.26

EX : 01

Basic Network Commands

Aim :

To Study and use basic networking commands for monitoring, troubleshooting and verify network connectivity.

Purpose :

* To Check network connectivity
* To Identify IP and MAC Addresses
* To Verify DNS resolution
* To analyze routing paths :

Command :

1) Ping :

Purpose - Checks whether a remote host is reachable and measure response time

Command - ping google.com

output - Reply from 142.250.183.14 : bytes = 32

time = 18ms TTL = 117

2) IP config :

Purpose : Display IP address, Subnet mask, and default gateway

Command : ipconfig

output : IPV4 Address :

192 . 168 . 1 . 10

3) ifconfig (Linux) :-

Purpose : Shows IP address and interface status

command : ifconfig

Output : inet 192 . 168 . 1 . 20 netmask

255 . 255 . 255 . 0

4) tracert (windows)

traceroute (Linux)

purpose : Identifies each router (hop) between source and destination.

Command : tracert google . com

output : 1 192 . 168 . 1 . 1

2 10 . 0 . 0 . 1

3 142 . 250 . 183 . 14

5) nslookup :

Purpose : Resolves a domain name into its IP address.

Command : nslookup google.com

output : Name : google.com
address : 142.250.183.14

6) net stat

purpose : Shows active connections, listening ports and routing information.

Command : TCP 0.0.0.0:80 Listening

7) arp

purpose : Maps IP address to MAC addresses

command : arp -a

output : 192.168.1.1 00-1a-2b-3c-4d-5e dynamic

8) hostname

purpose : Shows the system's network hostname

Command : hostname

Output - ABC123

9) route

Purpose : Shows network routes used for packet forwarding

Command : route print (windows)

Output : Network destination
0.0.0.0 Gateway
192.168.1.1

10) getmac (windows) :

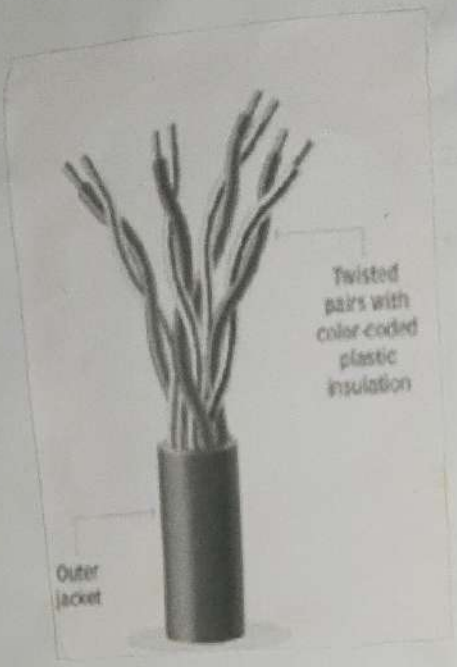
Purpose : Shows the physical address of network adapters

Command : getmac

Output : DD-1A-2B-3C-4D-5E

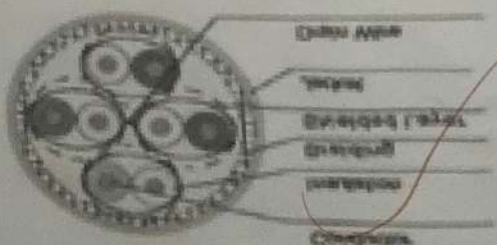
Result : The basic networking commands were executed successfully. Network connectivity, IP configuration, DNS resolutions and active network connections were observed and verified using the respective commands.

28/7/26





Shielded twisted pair



Aim:

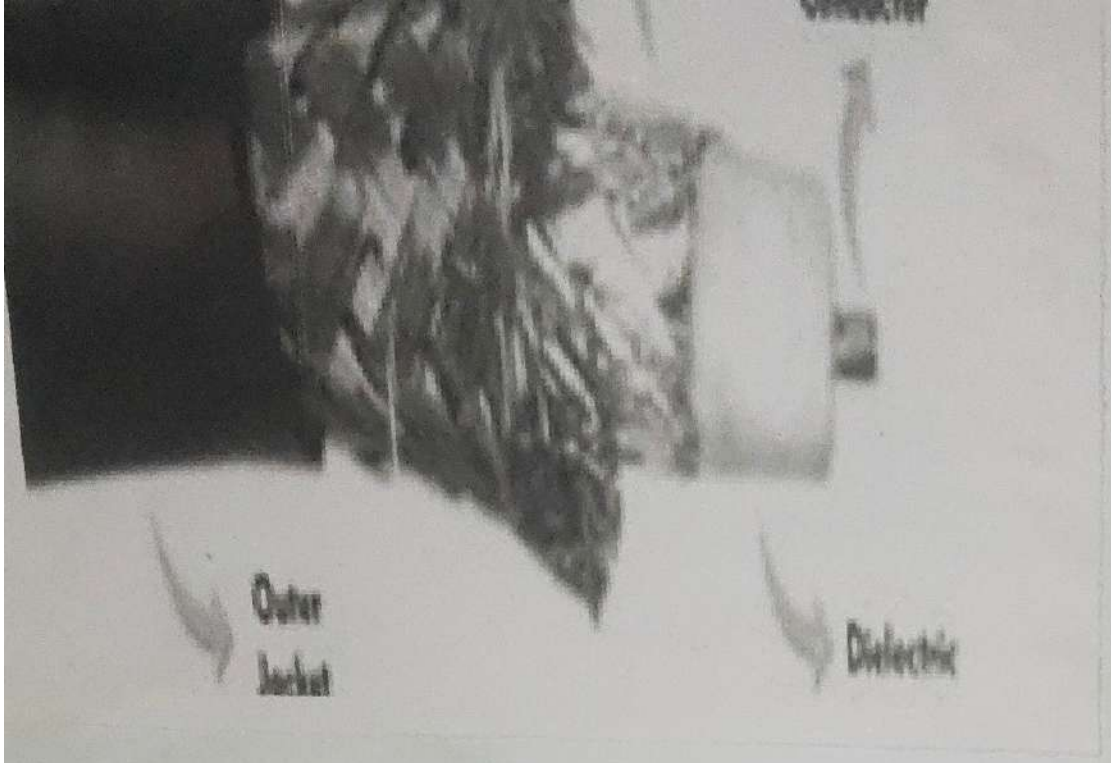
To study and understand different types of network cable, To make your own Ethernet Cross-over / Straight cable

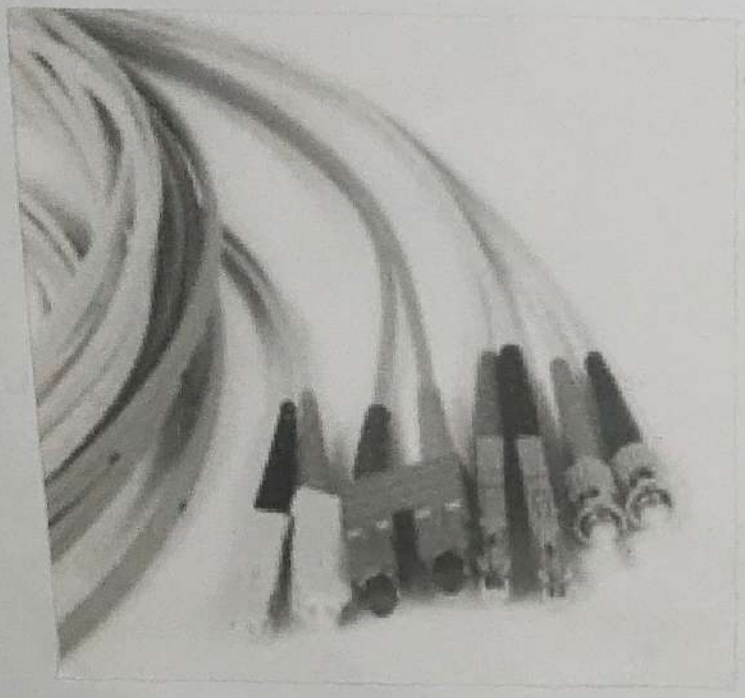
a) Understand different Study of network cable.

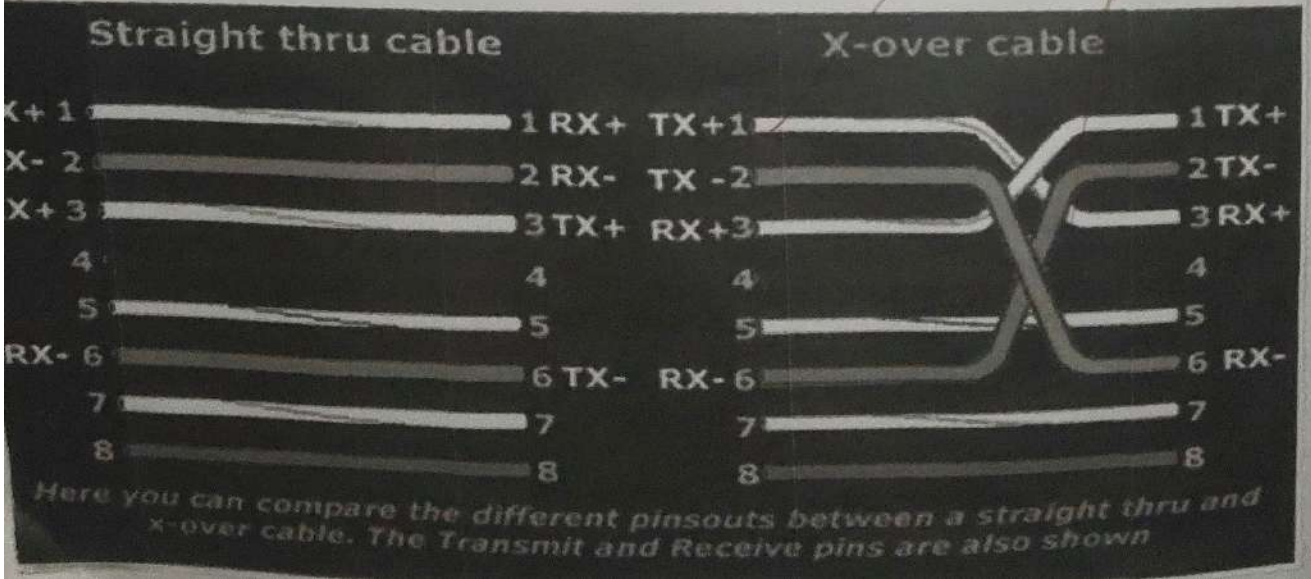
Different type of cables used in networking are:

1) Unshielded Twisted Pair (UTP) cable.
2) Shielded Twisted Pair (STP) cable
3) Coxial Cable
4) Fiber optic Cable.

Cable type	Category	Maximum Data Transmission	Advantage / Disadvantage	Application / use
UTP	Category 3	10 MBPS	Advantage: * Cheaper in cost * Easy to install as they have a smaller overall Disadvantage * More prone to (EMI) Electromagnetic	10base-T Ethernet
	Category 5	upto 100 MBPS		Fast Ethernet Gigabit Ethernet
	Category 5e	1GBPS		Fast Ethernet, Gigabit Ethernet







Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown

STP	Category 6, 6a	10 Gbps	Advantages * Shielded * Faster than UTP * Less susceptible to noise and interference Disadvantages * Expensive * Greater installation effort	Gigabit Ethernet, 10 Gb Ethernet (55m) widely used in data centers
SSTP	Category 7	10Gbps		Gigabit Ethernet, 10 Gb Ethernet (100m)
Coaxial Cable	RG-6 RG-59 RG-11	10-100MBPS	* High bandwidth * Immune to interference * Low loss Disadvantages * Limited distance * Cost	Speed of signal is 500m Television network High speed internet connections
Single mode Multimode	Fibre optics	100 Gbps	Advantages • High speed • High bandwidth • High security • Long distance Disadvantages • Expensive • Require skill	Maximum distance of fibre optic cable is around 1 centre


orange solid
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
Clip underneath, pins facing you
Clip underneath, pins facing you
green solid
brown sol
blue
white


A
B

Make your own Ethernet Cross-over Cable / Straight Cable.

Tools and parts needed:

* Ethernet cabling. CAT5e is certified for gigabit support
* A crimping tool. Run into the plug and cut the shielding off the cable
* Two RJ45 plugs

Step 1: To start construction of the device begin by threading Shields onto the cable

Step 2: Next; strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round area to compete this task

Step 3: After you will need to untangle the wires, there

Step 5 : Next, push the cable right in. The notch at the end of the plug needs to be just over the cable Shielding, and if it is not, that means that you stripped off to much Shielding.

Step 6 :- After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

Step 7 : Lastly, repeat for the other end using diagram B (to make a crossover cable) / using diagram A (to make a straight through cable)

Result :

(9.5)

Thus, different type of network cables were studied successfully., and Straight Through and cross over both Cables were prepared and tested Successfully

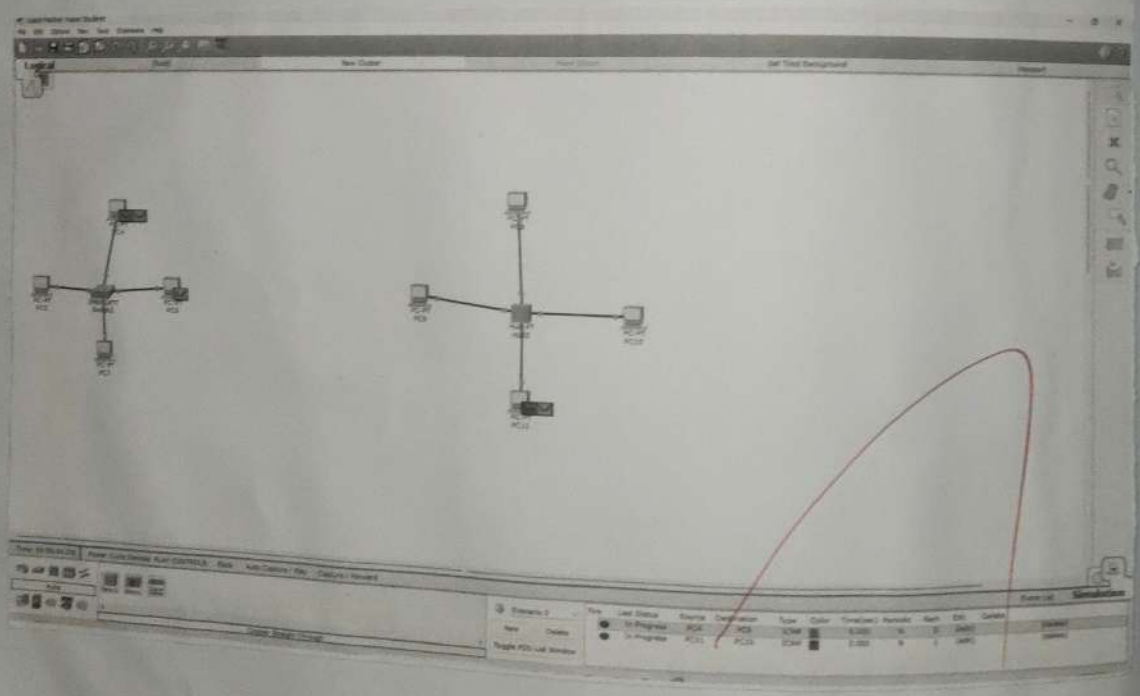
BM 6/8

Aim :

To understand the cisco packet tracer environment and using a simple computer network using its interface and networking components.

Steps :

1. Open the cisco packet tracer application
2. Observe the user interface, including the menu bar, Main Tool bar, work space, common tool bar, Network component Box, and Realtime Simulation tabs.
3. From the network Component Box, Select the required network devices such as pcs, Hub and Switch.
4. Drag and drop the devices into the work



PC using the desktop ⇒ IP configuration option

7. Verify that all the links are active

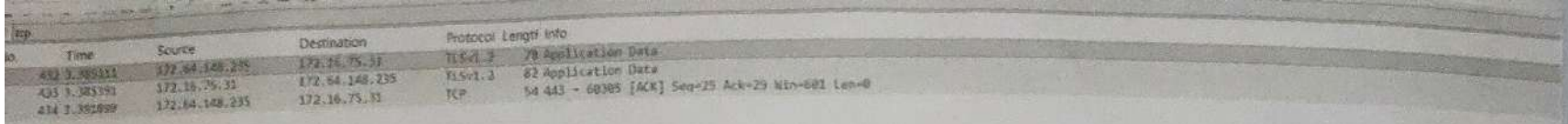
8. Use the add simple PDU tool to send a Packet between two PCs

9. Observe Packet transmission in Realtime mode and Simulation mode

10. Save the network topology.

Result:

The cisco packet tracer environment is successfully explored and a simple network was designed. The interface components, device connections, IP configuration, and Packet transmission were successfully understood using Realtime and Simulation mode.



Wireshark · Display Filters

Filter Name	Filter Expression
Ethernet address 00:00:5e:00:53:00	eth.addr == 00:00:5e:00:53:00
Ethernet type 0x0806 (ARP)	eth.type == 0x0806
Ethernet broadcast	eth.addr == ff:ff:ff:ff:ff:ff
No ARP	not arp
IPv4 only	ip
IPv4 address 192.0.2.1	ip.addr == 192.0.2.1
IPv4 address isn't 192.0.2.1	ip.addr != 192.0.2.1
IPv6 only	ipv6
IPv6 address 2001:db8::1	ipv6.addr == 2001:db8::1
TCP only	tcp
UDP only	udp
Non-DNS port	!(udp.port == 53 || tcp.port == 53)
TCP or UDP port is 80 (HTTP)	tcp.port == 80 || udp.port == 80
HTTP	http
No ARP and no DNS	not arp and not dns
Non-HTTP and non-SMTP to/from 192.0.2.1	ip.addr == 192.0.2.1 and tcp.port not in {80, 25}

HUB

Aim :-

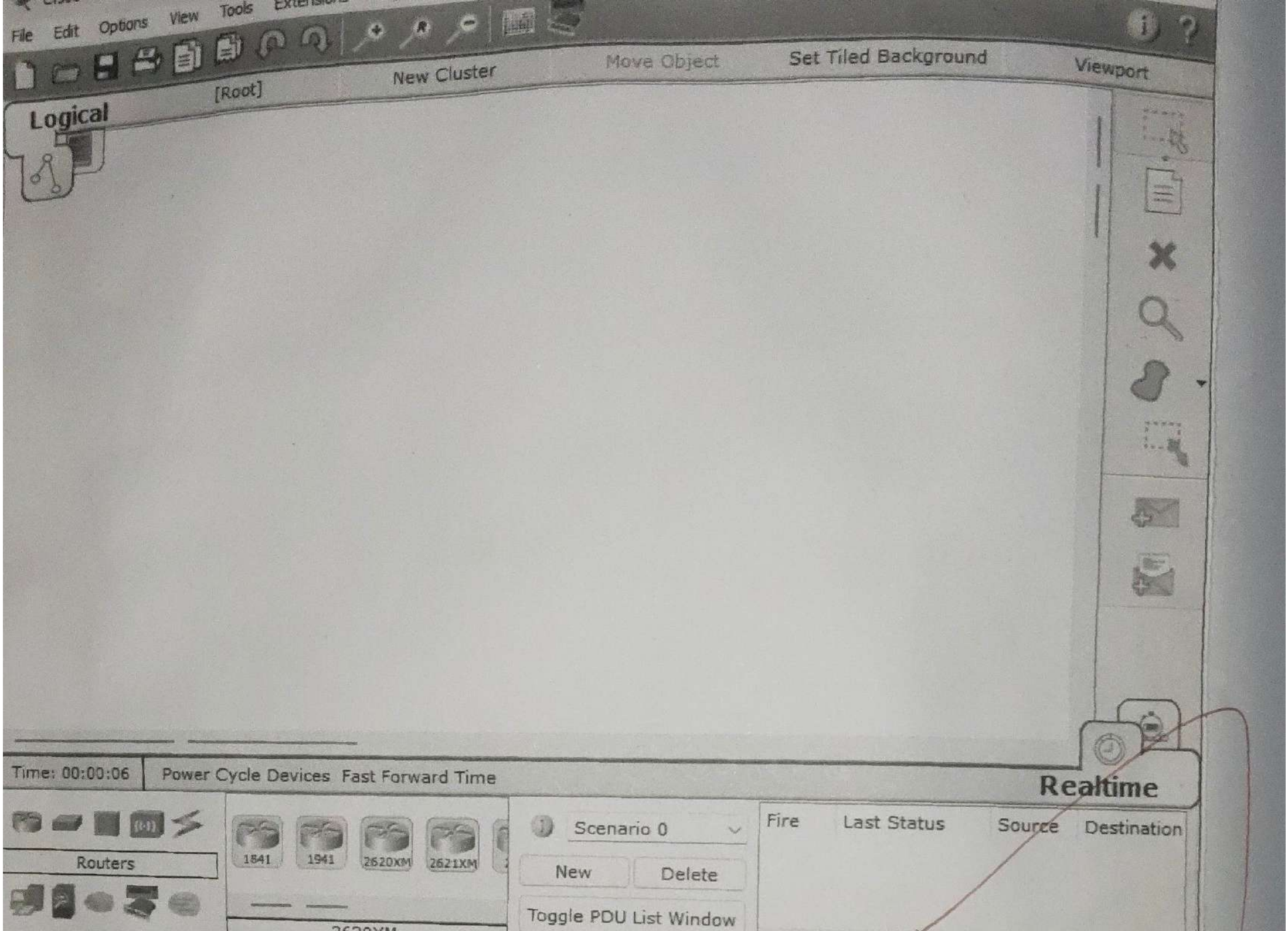
To study the behaviour of a hub by observing how it forwards data packets using Cisco Packet Tracer.

Steps :

1) Open Cisco Packet Tracer
2) Drag and place 1 hub and generic PCs with the workspace.
3) Connect all PCs to the Hub using copper straight - Through cable.
4) Configure IP address for all PC save network.
5) Select Add simple PDU from the common tools
6) Find a Packet from one PC to other
7) Switch to simulation mode and observe.

Result :

The hub broadcasts the received Packet to all connected devices. Every PC receives the Packet, but only the distination other discard it.


File Edit Options View Tools
[Root]
New Cluster
Move Object
Set Tiled Background
Viewport
Logical
Time: 00:00:06
Power Cycle Devices Fast Forward Time
Realtime
Routers
1841
1941
2620XM
2621XM
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination

Aim :

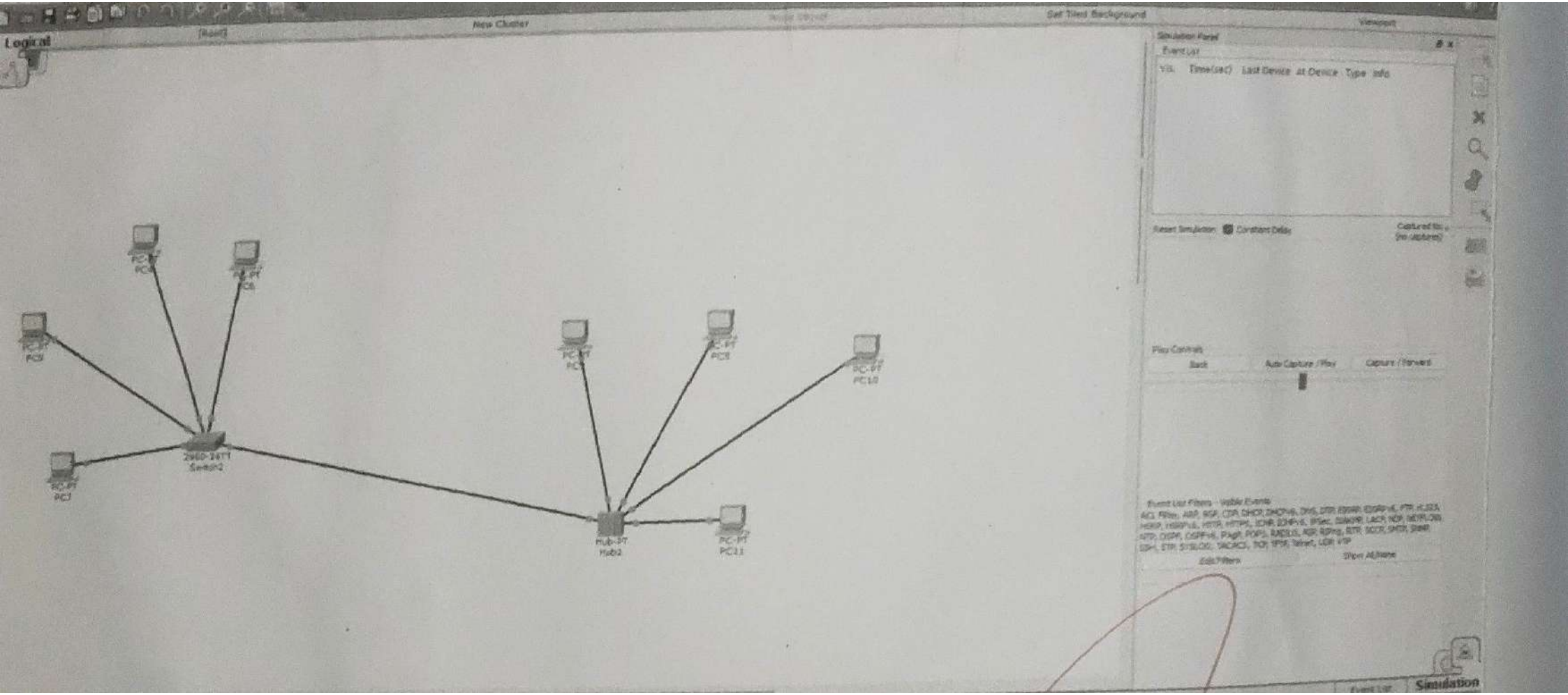
To study the behaviour of a switch by observing how it forwards data packets using cisco packet tracer.

Steps :

1) Open cisco packet tracer
2) Drag and Place 1 Switch and 4 Generic PCs into the workspace.
3) Connect all PCs to the Switch using Copper Straight Through cables.
4) Configure IP addresses for all PCs in the same
5) Select Add simple PDU from the another PC

Result :-

The switch forward the Packet only to the destination PC by using its MAC address table. This reduces unnecessary network traffic and improves efficiency


Logical
New Cluster
Set Tiled Background
Simulation Panel
Event List
Vis. Time(sec) Last Device At Device Type Info
Reset Simulation
Constant Delay
Play Controls
Back
Auto Capture / Play
Capture / Forward
Event List Filters - Visible Events
Edit Filters
Show All/None
Simulation

Aim :

To compare the Packet forwarding behaviour of a hub and switch using Cisco packet tracer.

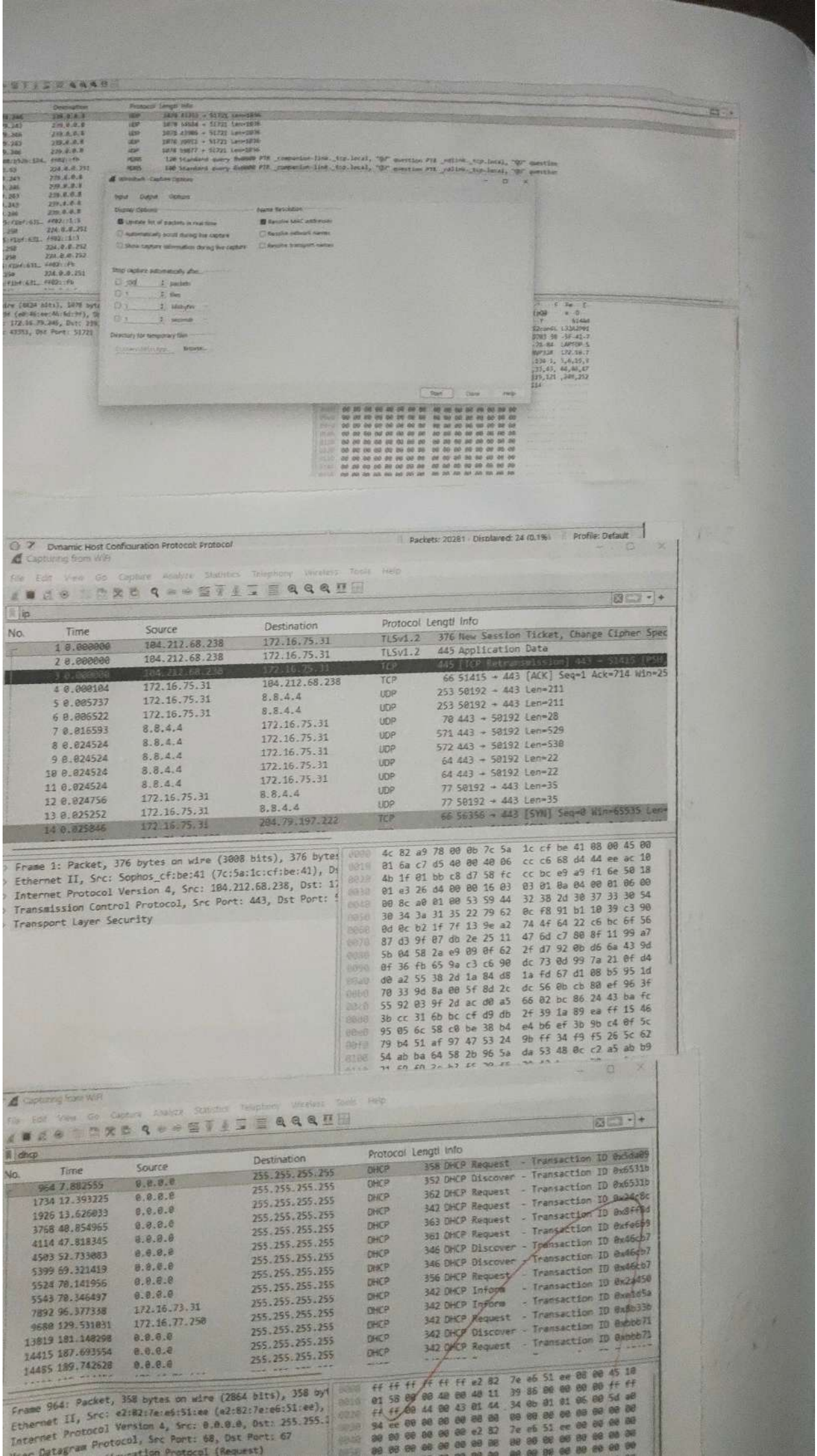
Steps :-

1) Create one network using a hub and another using a switch using cisco packet tracer.
2) Configure IP address for all PCs in both networks
3) Send a simple PDU between two PCs in each network.
4) Observe packet transmission in simulation mode.
5) Compare How the hub and switch forward packets

Result :

The hub broadcasts packets to all connected devices, whereas the switch forward packets only to the intended destination. Therefore, The switch provides better performance, reduced network traffic, and improved

6/8


Dynamic Host Configuration Protocol: Protocol
Packets: 20281 · Displayed: 24 (0.1%)
Profile: Default
Capturing from WiFi
ip
No. Time Source Destination Protocol Length Info
1 0.000000 104.212.68.238 172.16.75.31 TLSv1.2 376 New Session Ticket, Change Cipher Spec
2 0.000000 104.212.68.238 172.16.75.31 TLSv1.2 445 Application Data
3 0.000000 104.212.68.238 172.16.75.31 TCP 445 [TCP Retransmission] 443 → 51415 [PSH
4 0.000104 172.16.75.31 104.212.68.238 TCP 66 51415 → 443 [ACK] Seq=1 Ack=714 Win=25
5 0.005737 172.16.75.31 8.8.4.4 UDP 253 50192 → 443 Len=211
6 0.006522 172.16.75.31 8.8.4.4 UDP 253 50192 → 443 Len=211
7 0.016593 8.8.4.4 172.16.75.31 UDP 78 443 → 50192 Len=28
8 0.024524 8.8.4.4 172.16.75.31 UDP 571 443 → 50192 Len=529
9 0.024524 8.8.4.4 172.16.75.31 UDP 572 443 → 50192 Len=530
10 0.024524 8.8.4.4 172.16.75.31 UDP 64 443 → 50192 Len=22
11 0.024524 8.8.4.4 172.16.75.31 UDP 64 443 → 50192 Len=22
12 0.024756 172.16.75.31 8.8.4.4 UDP 77 50192 → 443 Len=35
13 0.025252 172.16.75.31 8.8.4.4 UDP 77 50192 → 443 Len=35
14 0.025846 172.16.75.31 204.79.197.222 TCP 66 56356 → 443 [SYN] Seq=0 Win=65535 Len=
Frame 1: Packet, 376 bytes on wire (3008 bits), 376 bytes
Ethernet II, Src: Sophos_cf:be:41 (7c:5a:1c:cf:be:41), Ds
Internet Protocol Version 4, Src: 104.212.68.238, Dst: 17
Transmission Control Protocol, Src Port: 443, Dst Port:
Transport Layer Security
dhcp
No. Time Source Destination Protocol Length Info
964 7.882559 0.0.0.0 255.255.255.255 DHCP 358 DHCP Request - Transaction ID 0x5da09
1734 12.393225 0.0.0.0 255.255.255.255 DHCP 352 DHCP Discover - Transaction ID 0x6531b
1926 13.626033 0.0.0.0 255.255.255.255 DHCP 362 DHCP Request - Transaction ID 0x6531b
3768 40.854965 0.0.0.0 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0x34c8c
4114 47.818345 0.0.0.0 255.255.255.255 DHCP 363 DHCP Request - Transaction ID 0x8ff8d
4503 52.733083 0.0.0.0 255.255.255.255 DHCP 361 DHCP Request - Transaction ID 0xfe669
5399 69.321419 0.0.0.0 255.255.255.255 DHCP 346 DHCP Discover - Transaction ID 0x46cb7
5524 70.141956 0.0.0.0 255.255.255.255 DHCP 346 DHCP Discover - Transaction ID 0x46cb7
5543 70.346497 0.0.0.0 255.255.255.255 DHCP 356 DHCP Request - Transaction ID 0x24450
7892 96.377338 172.16.73.31 255.255.255.255 DHCP 342 DHCP Inform - Transaction ID 0xe1d5a
9680 129.531031 172.16.77.250 255.255.255.255 DHCP 342 DHCP Inform - Transaction ID 0x8b33b
13819 181.140298 0.0.0.0 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0xbbb71
14415 187.693554 0.0.0.0 255.255.255.255 DHCP 342 DHCP Discover - Transaction ID 0xbbb71
14485 189.742628 0.0.0.0 255.255.255.255 DHCP 342 DHCP Request - Transaction ID
Frame 964: Packet, 358 bytes on wire (2864 bits), 358 byt
Ethernet II, Src: e2:82:7e:e6:51:ee (e2:82:7e:e6:51:ee),
Internet Protocol Version 4, Src: 0.0.0.0, Dst: 255.255.
User Datagram Protocol, Src Port: 68, Dst Port: 67
Dynamic Host Configuration Protocol (Request)

Experiment on Packet capture tool

Wireshark.

Aim:

To Study the features of wireshark as a packet capture and network analyser tool, and to capture, filter and analyse network packets at different protocol layers

Procedure:

1) Open the wireshark application on the computer
2) From the available network interfaces, select local area connection
3) Go to capture → options and select local area connection
4) Set the required network interface.
5) Click start to begin the capture network traffic
6) Generate some network activity by opening a website or performing operation
7) Observe the capture packets in the Packet list plane
8) Apply suitable display box to analyse specific protocols.

Result:

Ex :6
7/8/26

Setup and Configuration of Local Area Network using a Switch and Ethernet cable

Aim :

To setup and configure a Local Area network using a switch and Ethernet cable. assign IP address and verify communication between computers.

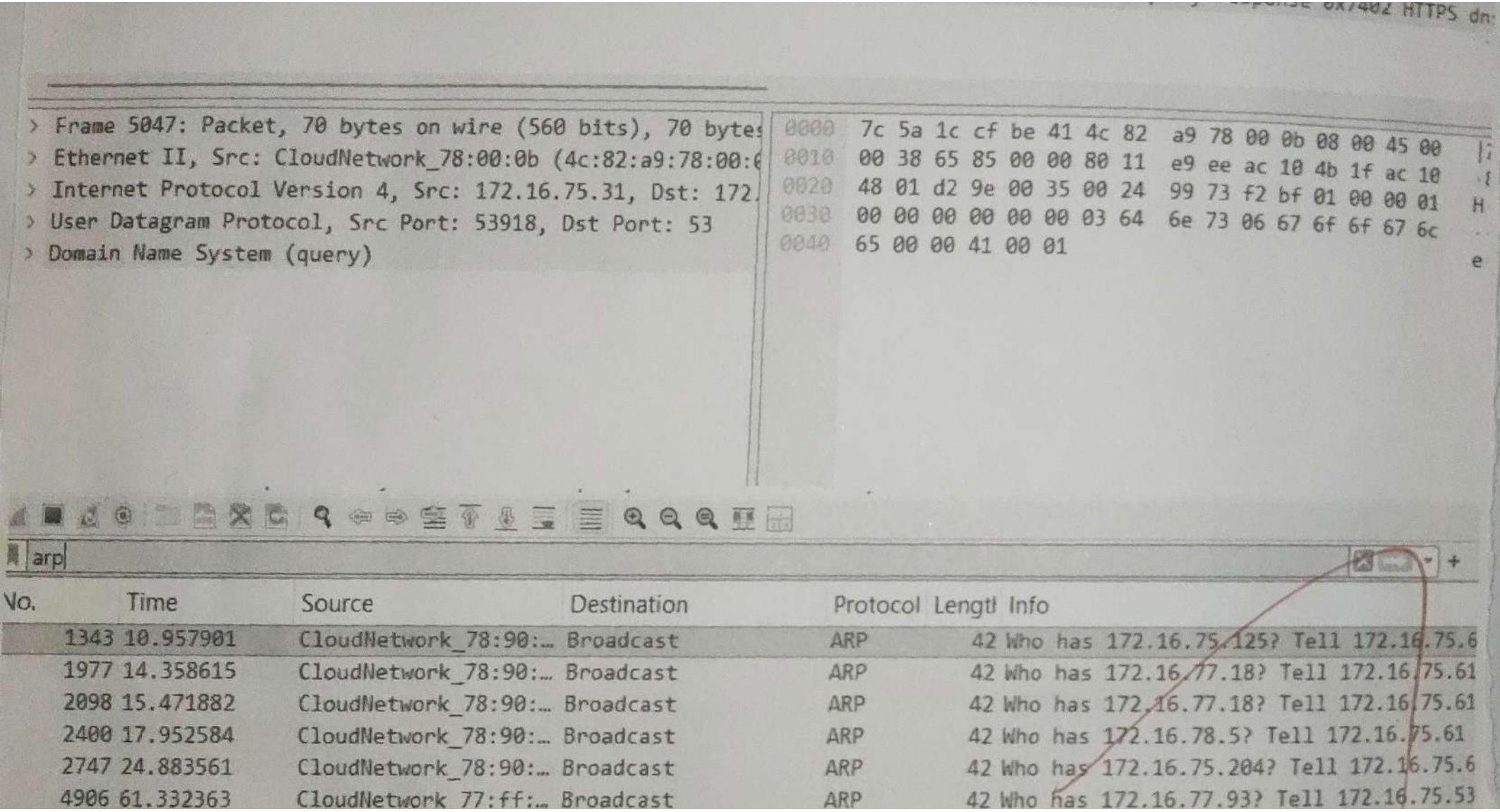
LAN :-

A local area network connects computers and devices with a small area such as a home, office, or school. It enables use the share files printers, internet access, and other resources.

Steps to Set up a LAN:

1) Plan and Design a appropriate network topology taking into account network requirement and equipment location.
2) Take 4 computers, a switch with 8, 16 or 24 which is sufficient for network.

Step 3: connect the Computers to network switch Ethernet cable, which is a simples as Plugging. one end of the Ethernet cable into the


0x7402 HTTPS dn
> Frame 5047: Packet, 70 bytes on wire (560 bits), 70 bytes
> Ethernet II, Src: CloudNetwork_78:00:0b (4c:82:a9:78:00:0
> Internet Protocol Version 4, Src: 172.16.75.31, Dst: 172.
> User Datagram Protocol, Src Port: 53918, Dst Port: 53
> Domain Name System (query)
0000 7c 5a 1c cf be 41 4c 82 a9 78 00 0b 08 00 45 00
0010 00 38 65 85 00 00 80 11 e9 ee ac 10 4b 1f ac 10
0020 48 01 d2 9e 00 35 00 24 99 73 f2 bf 01 00 00 01
0030 00 00 00 00 00 00 03 64 6e 73 06 67 6f 6f 67 6c
0040 65 00 00 41 00 01
arp
No. Time Source Destination Protocol Length Info
1343 10.957901 CloudNetwork_78:90:... Broadcast ARP 42 Who has 172.16.75.125? Tell 172.16.75.6
1977 14.358615 CloudNetwork_78:90:... Broadcast ARP 42 Who has 172.16.77.18? Tell 172.16.75.61
2098 15.471882 CloudNetwork_78:90:... Broadcast ARP 42 Who has 172.16.77.18? Tell 172.16.75.61
2400 17.952584 CloudNetwork_78:90:... Broadcast ARP 42 Who has 172.16.78.5? Tell 172.16.75.61
2747 24.883561 CloudNetwork_78:90:... Broadcast ARP 42 Who has 172.16.75.204? Tell 172.16.75.6
4906 61.332363 CloudNetwork_77:ff:... Broadcast ARP 42 Who has 172.16.77.93? Tell 172.16.75.53

Step : 8 :

Try to access the shared folders from other computers of the network.

Result :

A Lan was Successfully configured using a switch and Ethernet cable. IP address were assignment correctly, communication was Verified using the ping command Find file sharing was Successfully demonstrated.

88
Bill
bf

Error Detection and correction using Hamming code.

Aim:

To write a program to implement error detection and correction using Hamming code.

Algorithm:

1) Start the program
2) Read the number of 4-bit data blocks (n) forming the
3) For each 4-bit data blocks D = d1 d2 d3 d4, Perform Hamming encoding:
   a) Arrange 4 bits at positions 1, 2, and 4 and data bits at positions 3, 5, 6 and 7.
   b) Compute the parity bits using even parity:

      P1 = d1 XOR d2 XOR d4 (Covers bit positions 1, 3, 5, 7)
      P2 = d1 XOR d3 XOR d4 (Covers bit positions 2, 3, 6, 7)
      P4 = d2 XOR d3 XOR d4 (Covers bit positions 4, 5, 6, 7)

4) Simulate transmission by introducing a single bit error at a random bit position (entered by the user) in the code word.
5) At the receiver, recompute the check bits (syndrome) for
6) Compute Error position = C4 C2 C1 (binary number

d: ~~List~~ of [illegible]

h = [0]*7

h[2] = d[0]
h[4] = d[1]
h[5] = d[2]
h[6] = d[3]

h[0] = h[2] ^ h[4] ^ h[6]

h[1] = h[2] ^ h[5] ^ h[6]
~~h[2]~~ = h[4] *
h[3] = h[1] ^ h[5] ^ h[6]

return h

if detect_and_correct(h):

c1 = h[0] ^ h[2]
c2 = h[1] ^ h[2]
c4 = h[3] ^ h[1]

error_pos = c1 * 1 + c2 * 2 + c4

if error_pos ! 0.

```
=== HAMMING CODE - ERROR DETECTION AND CORRECTION ===

Enter number of 4-bit data blocks in the stream: 2

--- Block 1 ---
Enter 4 data bits (space separated, e.g. 1 0 1 1): 1 0 1 1
Original Data Bits                : 1011
Encoded Hamming Code (7 bits)     : 0110011
Enter bit position to introduce error (1-7, 0 = no error): 3
Transmitted Code (with error at position 3): 0100011
Received Code                     : 0100011
Checking for errors...
Error detected at bit position: 3 -> Correcting...
Corrected Code                    : 0110011
Recovered Data Bits               : 1011

--- Block 2 ---
Enter 4 data bits (space separated, e.g. 1 0 1 1): 0 1 1 0
Original Data Bits                : 0110
Encoded Hamming Code (7 bits)     : 1100110
Enter bit position to introduce error (1-7, 0 = no error): 6
Transmitted Code (with error at position 6): 1100100
Received Code                     : 1100100
Checking for errors...
Error detected at bit position: 6 -> Correcting...
Corrected Code                    : 1100110
Recovered Data Bits               : 0110
```

program to implement error det

amming code concept was succe

ed and verified using a sam

```
window_end = min(base + window_size - 1, total_frame - 1)

print("\n ----------")

print(f"Sender window : [Frame {base} - Frame {window_end}]")

while next_seq < base + window_size and next_seq < total_frame:

print(f"  [Frame {next_seq} lost in transit!]")

    next_seq += 1

if lost_frame is not None and base <= lost_frame < next_seq:

print(f"Node B --> Frame {lost_frame}")

print(f"Node A --> Timeout! Resending {lost_frame}")

next_seq = lost_frame

lost_frame = None

Continue.

(base, next_seq);

Send ACK {i}")
```

Flow Control at Dat

Sliding w

Aim :

To write a program to
Layer using the Sliding wi

Algorithm:

1) Start the program
2) Read the total number of f
Size (w), and a frame numbe
3) Initialize base = 0 and next
4) Repeat while base < N:
a) Determine the windo
b) Send all frames wi

main ()

```
=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go-Back-N) ===

Enter total number of frames to send: 8
Enter sender's window size: 4
Enter frame number to simulate as LOST (0 to 7, enter -1 for no loss): 5

-----------------------------------------
Sender Window : [Frame 0 - Frame 3]
Node A --> Sending Frame 0
Node A --> Sending Frame 1
Node A --> Sending Frame 2
Node A --> Sending Frame 3
Node B --> Received Frame 0, sending ACK 0
Node B --> Received Frame 1, sending ACK 1
Node B --> Received Frame 2, sending ACK 2
Node B --> Received Frame 3, sending ACK 3
Node A --> ACK 3 received. Sliding window forward.

-----------------------------------------
Sender Window : [Frame 4 - Frame 7]
Node A --> Sending Frame 4
Node A --> Sending Frame 5
   (Frame 5 LOST in transit!)
Node A --> Sending Frame 6
Node A --> Sending Frame 7
Node B --> Frame 5 missing. Discarding out-of-order frames after it.
Node A --> Timeout! Resending window starting from Frame 5.

-----------------------------------------
Sender Window : [Frame 4 - Frame 7]
Node A --> Sending Frame 5
Node A --> Sending Frame 6
Node A --> Sending Frame 7
Node B --> Received Frame 4, sending ACK 4
Node B --> Received Frame 5, sending ACK 5
Node B --> Received Frame 6, sending ACK 6
Node B --> Received Frame 7, sending ACK 7
Node A --> ACK 7 received. Sliding window forward.

-----------------------------------------
All 8 frames successfully transmitted from Node A to
```

number of hosts, assign static IP add
a router, and simulate packet transm
Cisco packet tracer.

Requirements:

* Cisco Packet Tracer Simulator
* Router
* Switches
* Personal Computer (PCs)
* Ethernet Cables
* Suitable IP address and subnet mask

Procedure:

i) open Cisco Packet Tracer and create a ne
network topology

2) Add one router, multiple switches and

* Switch to Simulation Mode in Cisco Packet Tracer
* Send a Simple PDU from a PC in one subnet to a PC
* Record the Result when the subnets are connected through the router.

Router Configuration

enable

Configure terminal

interface FastEthernet 0/0

ip address <IP address> <Subnet mask>

no shutdown

exit

ip address <IP address> <Subnet mask>

Switch mode acce

exit

Pc configuration:

Assign the required IP address, Subnet mask, default gateway and DNS server to each Pc.

Testing :-

Use the Ping- command from the PC Command prompt to verify connectivity between PCs and router.

Observation:

Subnetting was successfully implemented using a /27 subnet mask, and connectivity between the devices was verified using Ping.

Result:

Thus. Subnetting was successfully

Internet working using Router

Aim:

To design and configure a simple Internet using a router & two PCs in cisco packet tracer and test connectivity using PDU

Step 1:

Configure Router 1

1) Open Cisco Packet Tracer
2) Select Router → Router
3) open CLI and press enter
4) Enter: → Enable
   → Configure

Step : 2

· Configure PC 0 Router 2

IP address : 192 · 168 · 10 · 2

Subnet mask : 255 · 255 · 255 · 0

Default : 192 · 168 · 10 · 1

Gateway

Step 3: Connect the devices

Use copper ~~ethernet~~ 0 → Router 1

Fast ethernet 0/0

PC 0 Fast ethernet 0 → Router 1

Fast ethernet 0/0

Test connectivity

1) Select simulation mode
2) Choose add simple PDU
3) Click PC0 → PC1
4) If the PDU is successful, the two networks are communication through the router.

net was successfully designed and configured

of PC 0 PC was verified.

b] Wireless Router DHCP server and internet cloud.

Aim:

To design and configure an internetwork using a wireless router. DHCP servers and inter net cloud and verify connectivity.

Procedure:

Step 1: Build the network

Add: PC
wireless Router
Laptop
cable modem
Internet

Step 2: Connect Devices

Use the appropriate cables

=) PC Wireless Router (copper straight through

2) Select DHCP

3) The PC automatically receives IP address

Step 6: Configure Cisco services

DHCP

Go to Server → Service → DHCP

=) DHCP : ON

=) Pool name : DHCP Pool

=) Default gateway : 208 . 67 . 220 .

=) DNS Server : 208 . 67 . 220 . 220

=) Click add.

DNS

Go to Service → DNS

=) DNS : ON

=) Name : cisco.com

Step 7: Test connectivity

1) Open PC → Desktop → command prompt

2) Type : ip config / release

3) Then ipconfig / renew

4) Check that the PC gets a 192 . 168 .

Result: internet work was successful

2) Accept client connection.

3) Creation a seperate thread

4) Receive message from the cl

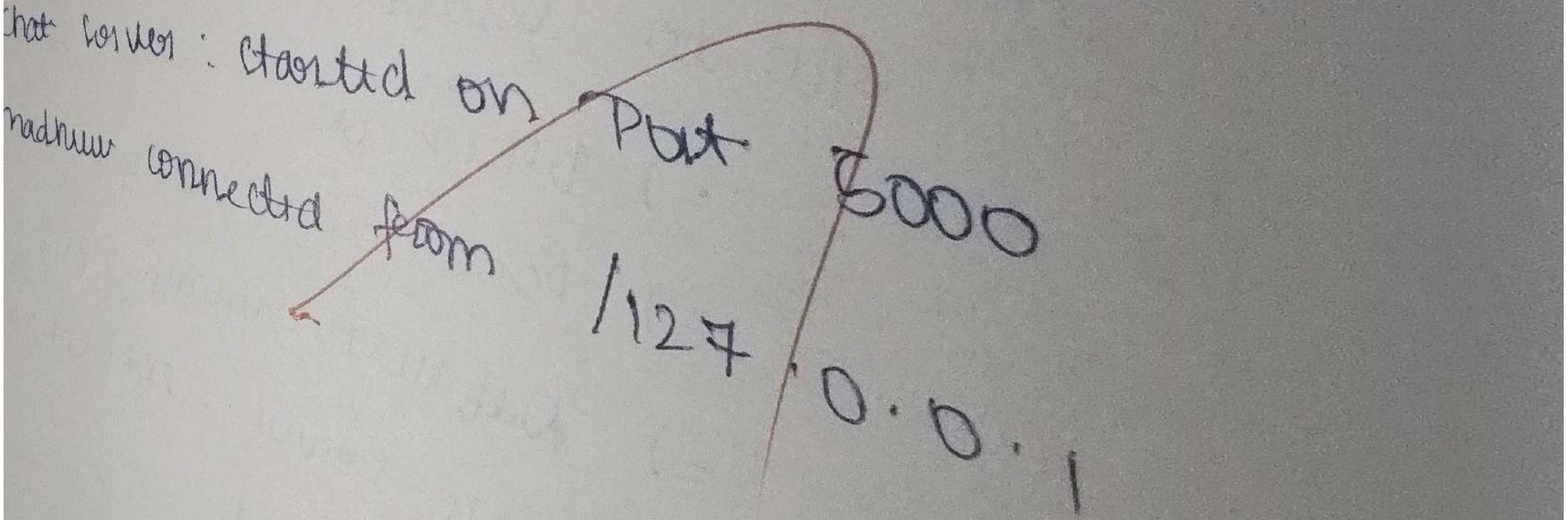
5) Broadcast the message to all

6) Remove the client when quit i

7) Close the communicaton.

ode: Server:

```
import java.io.*;
import java.net.*;
import java.util.*;
Public class Chatserver {
    private static final int Port = 5000
    private static final set < client
        Handler > clients = collection.
```


Chat server : Started on Port 6000
Madhuu connected from /127.0.0.1

```
private static final String server_address
        = "127.0.10.1";
public static void main (String [] args)
    throws IOException {
        Socket socket = new Socket (server
        address, server_port);
    Thread listener = new Thread () -> {
    try {
        while ((server Message = in.readline() ) != null){
        System.out.println (" Dis connected
                from server " );
    }
};
listen.setDaemon (true);
listen.start ();
String UserInput;
while ((UserInput = console.readline () ) !=
                NULL){
        out.println (In (UserInput));
    }
    Socket.close ();
}
}
```

Result:

Thus the Chat Server and Chat Client were successfully connected and message were

7/26
12a)

Aim:

Implement echo client server using TCP / UDP sockets

Algorithm:

=) Start the client and set the server host & port

=) Create a TCP socket & connect to the server

=) Read a message from the user

=) Send the message to the echo server

=) Recieve and display the server response

=) Close the socket & end the connection

code:

TCP echo client :-

```
import java.io.*;
import java.net.*;
public class TCPEchoClient {
```

Prac 1:-

TCP echo server listening on port 5000

TCP echo server listening on port 5000

```
                quit ) : ");

        String userInput;

            while ((userInput = stdIn.readLine()
                null ){ out.println (userInput

        String response = in.readLine();

        System.out.println ("bye")) {

            break;

        }

    }

}

System.out.println("Connection Closed");

}

}
```

TCP Echo server

```
import java.IO.*;

import java.net.*;

public class TCPEchoServer {

    public static void main (String[
```

local host : 6000

Type message to send (type bye to quit)

hello

Server : Echo : hello

udp

Server : Echo : udp

```
Reader in new BufferedReader in
Reader (InputStreamReader());
String line;
while ((line = in.readLine()) != null){
    System.out.println("Recieved : "+line);
}
}
System.out.println("Client disconnected");
```

Result :

The TCP echo client and UDP echo client server
TCP / UDP Sockets implemented.

Routing at Network layer

a) simulate static routing protocal configuration using Cisco Packet tracer

Aim:

To stimulate and configure static routing between multiple networks using Cisco Packet Tracer, configure primary and backup static routes

Requirements :-

=) Cisco Packet Tracer
=) Three router - Router 0, Router and Router 2
=) PCs / end devices
=) Ethernet / serial connection as required.

Observation :-

=) A network topology consisting of 3 routers, Router 0, Router 1 and Router 2 was created in Cisco Packet Tracer. Different networks were connected to the routers.

Router 1 :

Router 1 was configured with static routes to the 10.0.0.0/8 and 40.0.0.0/8 networks the route through router 0 was assigned a lower administrative distance of 10. The alternate

Verified using Ping and tracert, confirming that backup route was working.

Verification :-

Show ip route static

The routing table showed the configuration static route and their administration distance values..

Deleting Static route :-

Show ip route static.

After deleting the main route, the backup route, the backup route became the active route if the primary path is

were configured for the network topology was created in cisco Packet tracer using Router 0, Router 1 and Router 2. The required IP address were configured on the router interface and PCs.

Router 1 :-

=) Router 1 was configured with static router to 10.0.0.0/8 and 10.0.0.0/8. The route through router 0 was given AD/0 as the main route, while the alternate route was Verified.

Router 2:

router 2 was configured with static router to 10.0.0.0/8 through router and 30.0.0.8/ through router 1. Both routes are sucessfully added to the routing table and Verified.

Static Verification :-

=) The configured static routes were checked using the show ip route static. connectivity was verified using Ping and traced.

result confirmed that the backup route was working correctly.

Result :

The static routing was successfully simulated and configured using cisco packet - tracer. The main and backup route were successfully configured.

Ex No: 13
16/06

Implement your own Ping Program.

Aim:-

To implement a simple ping program using socket programming to check the connectivity & response time between two network devices.

Algorithm:-

=> Start the program.
=> Get the destination IP address / hostname from the user.
=> Creat a socket for sending & receiving packets.
=> Set a socket from the destination.
=> If a response is received, calculate & display the output.
=> Repeat the process for the required number of Packets
=> Stop the Packet.

Program:-

```
import socket
import time
host = input ("Enter destination IP address:")
count = int (input ("Enter number of Packets:"))
for i in range (count):
    s = socket (socket.AF_INET)
```

Enter destination IP address : 142.250.123.14

Enter number of Packet : 4

Reply from 142.250.123.4 time = 25.31 ms.

Reply from 142.250.183.14 : time = 24.52ms

Reply from 142.250.183.14 : time = 26.14ms

Ex.No: 14
Date:26

Write a code using raw sockets to implement Packet Sniffing

Aim:

To implement a basic packet sniffer using raw socket & display network packet information.

Algorithm:-

⇒ Start the program

⇒ Create a raw socket

⇒ Bind the socket to the local network interface

⇒ Recieve packet from network

⇒ Display the source MAC address destination Mac address and Protocal.

⇒ Repeat for incoming packets.

⇒ Stop the program.

Program:-

```
import socket
import struct
hostname = socket.gethostname()
ipaddress = socket.gethostbyname(hostname)
print("Packet Sniffer")
print("Host name:", hostname)
```

```
print ("Packet Sniffer")
print ("Host name :", host name)
print ("IP address :" ip_address)
sock = socket.socket (socket.AF_INET, socket)
sock.setsockopt (socket.IPPROTO_IP socket, IP_HDRINCL
print ("Sniffing packet...")
try:
    count = 0
    while count < 5
    packet, address = sock.recv from (65535)
    print ("Source IP :" source_IP)
    print ("Destination IP :", destination_IP)
    print ("protocal :", protocal)
except Keyboard interrupt
    print ("\n packet sniffing stopped:")
finally:
```

output :-

Packet sniffer

Hostname : Desktop-ABC123

IP address : 192.168.05

sniffing packets

source IP : 192.168.05

Destination IP : 192.168.0.1

protocol : 6

packet size : 60 bytes

source IP : 192.168.0.5

protocol : 6

Packet size : 60 bytes

Result :-

The packet sniffer successfully captured network packet and displayed the source IP, destination IP

8

Wireless LAN.

Aim:

To configure a wireless LAN using a linksys wireless Router in in cisco Packet tracer and securely connect three PCs using a WEP Security Key.

Observation :-

⇒ A Linksys wireless Router was connected to three PCs to find a wireless LAN

⇒ The router IP address was configured as 192.168.

⇒ The SSID of the wireless network was confirmed as MotherNetwork

⇒ WEP security was enabled on the wireless router with the security key 0123467 89

Result:

Thus the Wireless LAN was successfully configured using a linksys wireless Router in ciscoPacket tracer and all 3 were

Exp no:5
7/10/26

Analyze Various types of servers

Webalizer tool.

Aim:-

To analyze the web server log files and generate statistical reports using the webalizer tool.

Observation:-

1) Webalizer is a web server log analysis tool it used to analyze server activity

2) The web server log file is given as input to webalizer such as

- ⇒ Number of visits
- ⇒ Number of pages
- ⇒ Number of pages
- ⇒ Data transfer.
- ⇒ The analyzed information is displayed as HTML reports, code commands.

Webalizer is mainly used through Linux commands rather than a programming code.

Sample output: Webalizer version 2.23-08

Generating report for

Apache Web Server

Summary: Total hits : 1250

Total files : 950

Total Pages : 650

Total visits : 185